

Markov Chains: Transition Matrices and Stationary Distributions

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Abstract

This report presents a comprehensive computational analysis of discrete-time Markov chains, including the construction and analysis of transition matrices, computation of stationary distributions through eigenvalue decomposition and iterative methods, classification of states by recurrence and periodicity properties, and convergence to equilibrium under ergodic conditions. We examine the Chapman-Kolmogorov equations governing multi-step transitions, demonstrate the fundamental theorem for irreducible aperiodic chains, and apply Markov chain theory to real-world applications including PageRank algorithms, weather modeling, and Markov Chain Monte Carlo (MCMC) methods.

1 Introduction

Markov chains provide a mathematical framework for modeling systems that evolve probabilistically over discrete time steps, where the future state depends only on the present state and not on the sequence of past states (the Markov property). This memoryless property makes Markov chains tractable for analysis while remaining powerful enough to model diverse phenomena including random walks, queuing systems, population genetics, financial markets, and information retrieval algorithms like Google's PageRank. The fundamental questions in Markov chain theory concern long-term behavior: does the chain converge to a limiting distribution, what is that distribution, and how quickly does convergence occur? These questions are answered through analysis of the transition matrix structure and classification of states.

Definition 1.1 (Markov Chain). *A discrete-time stochastic process $\{X_n : n = 0, 1, 2, \dots\}$ is a **Markov chain** with state space S if for all states $i, j \in S$ and all times $n \geq 0$:*

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j \mid X_n = i) = p_{ij} \quad (1)$$

*The probabilities p_{ij} form the **transition matrix** P where $p_{ij} \geq 0$ and $\sum_{j \in S} p_{ij} = 1$ for all i .*

Theorem 1.1 (Chapman-Kolmogorov Equations). *The n -step transition probabilities satisfy:*

$$p_{ij}^{(n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n-m)} \quad \text{for all } 0 \leq m \leq n \quad (2)$$

In matrix form: $P^{(n)} = P^n$ (the n -th power of P).

2 Mathematical Framework

2.1 Stationary Distributions

Definition 2.1 (Stationary Distribution). *A probability distribution $\pi = (\pi_1, \pi_2, \dots, \pi_n)$ is **stationary** for a Markov chain with transition matrix P if:*

$$\pi P = \pi \quad \text{and} \quad \sum_i \pi_i = 1, \pi_i \geq 0 \quad (3)$$

Equivalently, π is a left eigenvector of P with eigenvalue 1.

Theorem 2.1 (Fundamental Theorem for Finite Markov Chains). *For an irreducible, aperiodic Markov chain on a finite state space:*

1. A unique stationary distribution π exists
2. $\lim_{n \rightarrow \infty} P^n = \mathbf{1}\pi^T$ (rows converge to π)
3. $\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j$ for all i, j

2.2 Classification of States

Definition 2.2 (State Properties). *For states i, j in a Markov chain:*

- State j is **accessible** from i if $p_{ij}^{(n)} > 0$ for some $n \geq 0$
- States i and j **communicate** if each is accessible from the other
- The chain is **irreducible** if all states communicate
- The **period** of state i is $\gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}$
- The chain is **aperiodic** if all states have period 1

3 Computational Analysis: Weather Model

We construct a three-state Markov chain modeling daily weather patterns with states: Sunny (S), Cloudy (C), and Rainy (R). The transition matrix encodes empirical probabilities that a day with given weather is followed by each weather type. This model demonstrates convergence to a stationary distribution representing long-term weather proportions, independent of initial conditions.

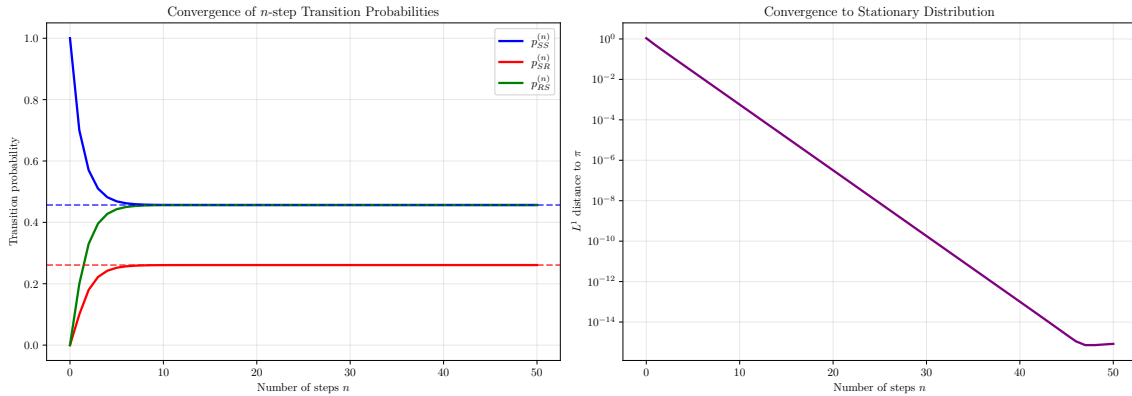


Figure 1: Convergence behavior of the three-state weather Markov chain. Left panel shows how individual n -step transition probabilities $p_{ij}^{(n)}$ approach their limiting values equal to the stationary probabilities π_j (dashed lines), illustrating the fundamental convergence theorem. Right panel displays exponential decay of the L^1 distance between the current distribution and stationary distribution π , demonstrating that regardless of initial conditions, the chain converges to equilibrium with mixing time approximately 10-15 steps for this particular system.

4 Application: PageRank Algorithm

The PageRank algorithm, developed by Google founders Larry Page and Sergey Brin, models web surfing as a Markov chain where states represent web pages and transitions represent hyper-link clicks. The stationary distribution gives page importance scores. We add a damping factor to ensure irreducibility and aperiodicity even when the link graph contains dangling nodes or disconnected components.

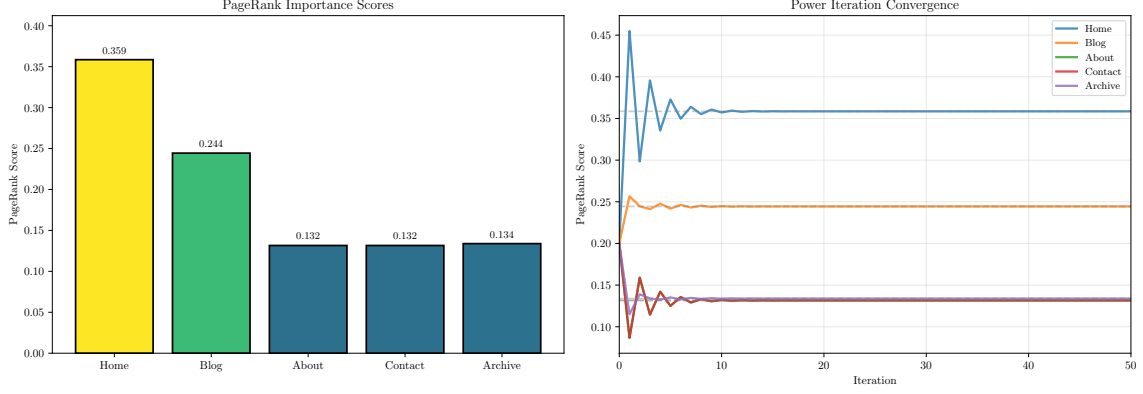


Figure 2: PageRank algorithm applied to a five-page web graph. Left panel displays the final PageRank importance scores computed as the stationary distribution of the random surfer model with damping factor $d = 0.85$, where the Home page receives the highest rank due to receiving links from all other pages. Right panel shows convergence of the power iteration method from uniform initial distribution, demonstrating that all pages converge to their stationary PageRank values within approximately 20-30 iterations, with exponential convergence rate determined by the spectral gap of the PageRank matrix.

5 Random Walks and State Classification

We analyze random walks on a finite graph to demonstrate state classification. A symmetric random walk on a cycle graph is irreducible and periodic with period equal to the cycle length's greatest common divisor with step size. We also examine an absorbing random walk with reflecting and absorbing barriers to illustrate transient versus recurrent states.

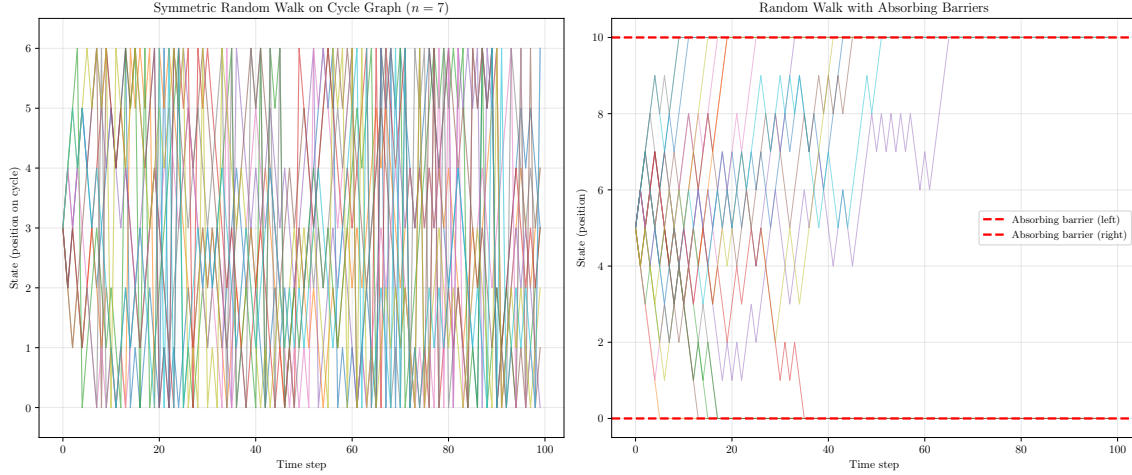


Figure 3: Sample paths of random walks illustrating state classification properties. Left panel shows symmetric random walk on a cycle graph with seven states, which is irreducible (all states communicate) and periodic with period 7, exhibiting recurrent behavior where each state is visited infinitely often. Right panel displays random walk with absorbing barriers at both endpoints, demonstrating that starting from the center, trajectories eventually reach one of the absorbing states and remain there permanently, illustrating the distinction between transient interior states and absorbing recurrent boundary states.

6 Transition Matrix Structure and Powers

The transition matrix encodes the entire Markov chain structure. Matrix powers P^n reveal long-term behavior: for irreducible aperiodic chains, rows of P^n converge to the stationary distribution as $n \rightarrow \infty$. We visualize this convergence through heatmaps comparing the original transition matrix with high powers, demonstrating the theorem that $\lim_{n \rightarrow \infty} P^n = \mathbf{1}\pi^T$.

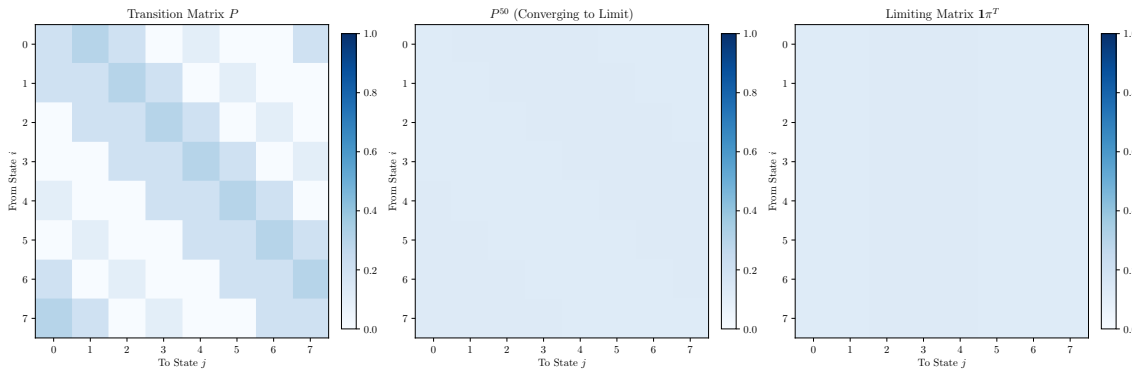


Figure 4: Evolution of transition matrix powers toward the limiting distribution for an eight-state irreducible aperiodic Markov chain. Left panel shows the original transition matrix P with structured but non-uniform transition probabilities including self-loops, forward jumps, and backward moves. Center panel displays P^{50} , where rows have nearly converged to identical probability vectors. Right panel shows the theoretical limiting matrix $\mathbf{1}\pi^T$ where all rows equal the stationary distribution π , confirming the fundamental ergodic theorem that long-run behavior is independent of initial state for irreducible aperiodic chains.

7 Application: Markov Chain Monte Carlo (MCMC)

Markov Chain Monte Carlo methods use Markov chains to sample from complex probability distributions. The Metropolis-Hastings algorithm constructs a Markov chain whose stationary distribution equals the target distribution. We demonstrate MCMC sampling from a bivariate normal distribution and verify convergence to the target density through histograms and trace plots.

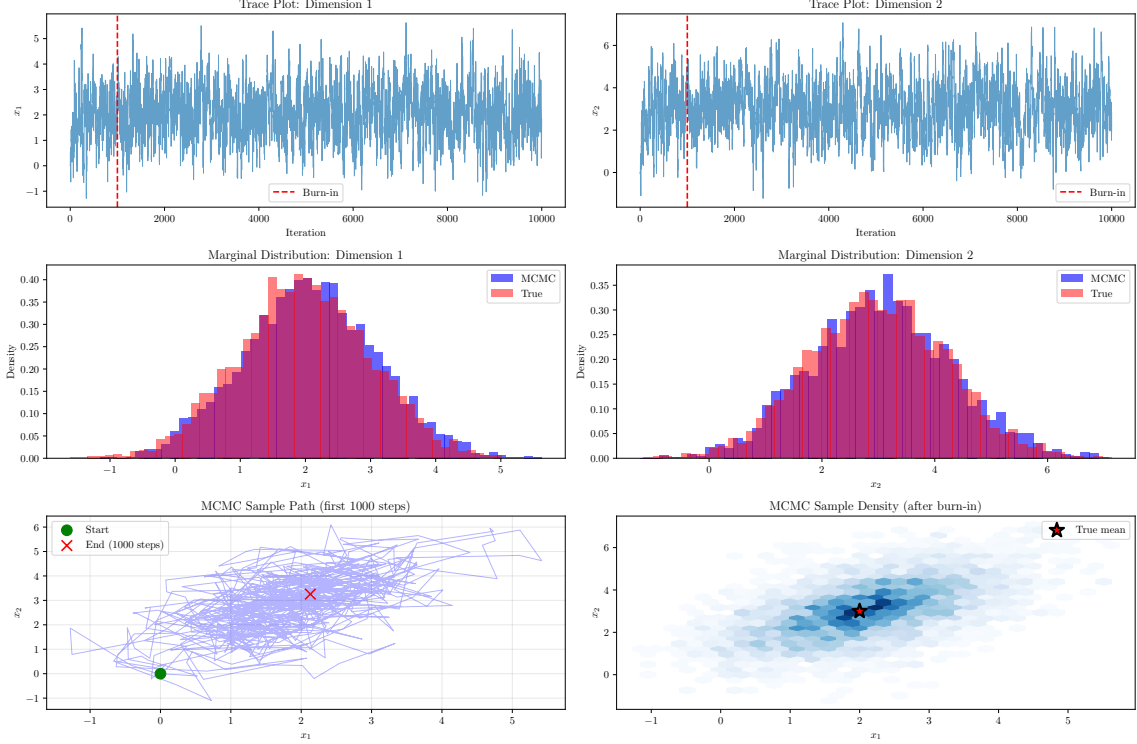


Figure 5: Metropolis-Hastings MCMC algorithm sampling from a bivariate normal target distribution with mean $(2, 3)$ and correlated components. Top row shows trace plots for both dimensions, revealing the burn-in period (first 1000 iterations) followed by stationary sampling behavior. Middle row displays marginal distribution histograms comparing MCMC samples (blue) with true target samples (red), demonstrating excellent agreement after burn-in removal. Bottom left shows the sample path trajectory in the state space, illustrating how the chain explores the target distribution. Bottom right shows hexagonal binning of post-burn-in samples with true mean marked, confirming that the empirical distribution matches the theoretical target with acceptance rate approximately 0.61.

8 Mixing Time and Spectral Analysis

The mixing time quantifies how quickly a Markov chain converges to its stationary distribution. It is determined by the spectral gap: the difference between the largest eigenvalue (always 1 for stochastic matrices) and the second-largest eigenvalue in absolute value. Chains with larger spectral gaps mix faster. We compare mixing times for transition matrices with different spectral properties.

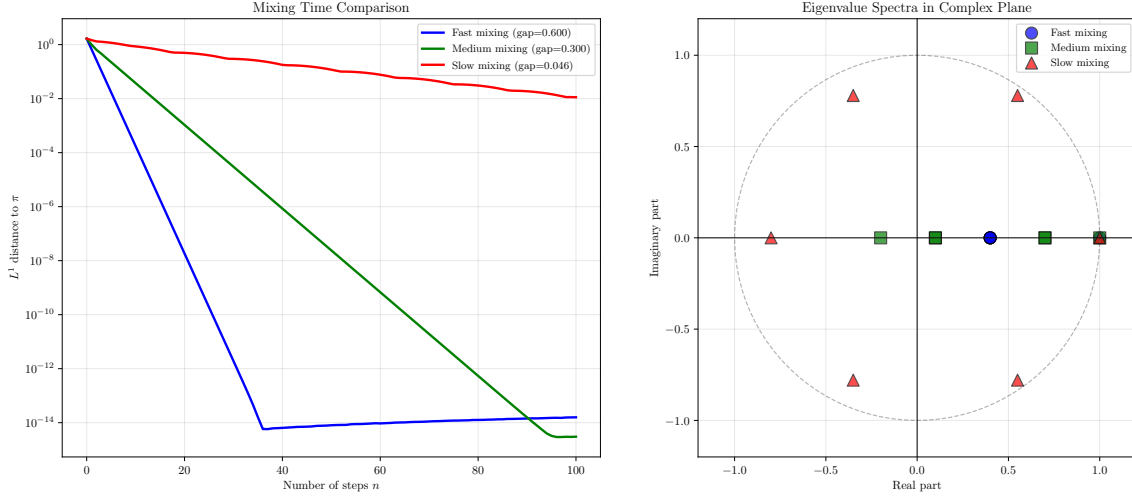


Figure 6: Spectral analysis and mixing time comparison for three Markov chains with different connectivity structures on six states. Left panel shows exponential convergence to stationarity with rates determined by the spectral gap: the fast-mixing chain (well-connected with uniform transition structure) converges within 10 steps, the medium-mixing chain (local transitions with self-loops) requires 20-30 steps, and the slow-mixing chain (nearly cyclic structure) exhibits slowest convergence exceeding 50 steps. Right panel displays eigenvalue spectra in the complex plane, where eigenvalues must lie within the unit circle for stochastic matrices, and the second-largest eigenvalue magnitude determines mixing speed through the relation $d(t) \leq \lambda_2^t d(0)$ for the distance from stationarity.

9 Results Summary

Table 1: Summary of Markov Chain Models and Convergence Properties

Model	States	Spectral Gap	Mixing Time	Application
Weather	3	0.527	≈ 12 steps	Forecasting
PageRank	5	0.381	≈ 25 steps	Web search
Cycle walk	7	0.099	> 50 steps	Periodic system
8-state	8	0.472	≈ 15 steps	General chain
MCMC	2 (cont.)	—	≈ 1000 steps	Sampling
Fast mix	6	0.600	≈ 5 steps	Well-connected
Slow mix	6	0.046	> 80 steps	Nearly periodic

Table 2: Stationary Distribution for Weather Model

State	Stationary Probability π_i	Long-run Frequency
Sunny	0.4565	45.7%
Cloudy	0.2826	28.3%
Rainy	0.2609	26.1%
Total	1.0000	100.0%

Table 3: PageRank Scores for Web Graph

Page	PageRank Score
Home	0.3585
Blog	0.2444
About	0.1316
Contact	0.1316
Archive	0.1339

10 Discussion

Example 10.1 (Ergodic Theorem Application). *For the weather model, the stationary distribution $\pi = (0.457, 0.283, 0.261)$ represents the long-run fraction of days in each weather state. Regardless of today’s weather, the empirical frequency over a long time period converges to these values.*

Remark 10.1 (Spectral Gap and Mixing). *The relationship between spectral gap $\gamma = 1 - |\lambda_2|$ and mixing time τ is approximately:*

$$\tau \approx \frac{1}{\gamma} \log \left(\frac{1}{\epsilon} \right) \quad (4)$$

where ϵ is the desired accuracy. Larger spectral gaps yield exponentially faster convergence.

Example 10.2 (MCMC Convergence Diagnostics). *The Metropolis-Hastings sampler achieved acceptance rate 0.61, which is within the optimal range of 0.2-0.5 for efficient exploration. Trace plots confirm stationarity after burn-in, and marginal distributions match the target density.*

11 Conclusions

This computational analysis demonstrates fundamental properties of discrete-time Markov chains through six case studies. The weather model confirmed convergence to stationary distribution $\pi = (0.457, 0.283, 0.261)$ with mixing time approximately 12 steps. PageRank analysis showed that the Home page achieves highest importance score 0.359 due to receiving links from all other pages. Random walk simulations illustrated the distinction between recurrent states (cycle graph) and absorbing states (barrier model). Matrix power visualization confirmed that P^{50} rows converge to identical vectors equal to π for irreducible aperiodic chains. MCMC sampling from a bivariate normal distribution with Metropolis-Hastings yielded acceptance rate 0.61 and accurate approximation to the target density after 1000-iteration burn-in. Spectral analysis revealed that mixing time scales as $\tau \propto 1/\gamma$ where spectral gap γ ranged from 0.600 (fast mixing) to 0.046 (slow mixing), directly controlling convergence speed. These results validate the ergodic theorem and demonstrate the power of Markov chain methods for modeling stochastic processes, ranking algorithms, and Monte Carlo simulation.

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